

Working with Virtual Serial Ports

Requisitos de conclusão

This page explains how to configure and use virtual serial ports on Linux, in the context of the Lab 1 protocol work.

The Lab 1 protocol work can be developed outside the laboratory using virtual serial ports in a single PC. This strategy does not eliminate the need to test the protocol in the laboratory with real serial ports, since there are small differences between using virtual and real hardware.

Prerequisites

This guide was written for using virtual serial ports in Linux.

If you are using Windows, you can use [Windows Subsystem for Linux 2 \(WSL2\)](#) as an alternative to creating virtual machines with Linux. It is faster and uses fewer resources than virtual machines.

If you prefer to use virtual machines, you can create them with [VirtualBox](#) or [VMware Player](#). Read the section below about using virtual machines and virtual serial ports.

cable.c Program (Recommended)

The cable.c program (available in the code examples folder in Moodle) is a program written in C that creates two virtual serial ports in your PC. The two virtual serial ports are connected to each other and allow sending bytes through one virtual serial port and receiving them in the other virtual serial port.

It is recommended to use the cable.c program when working with virtual serial ports in your PC. This program allows students to work on the protocol outside the laboratory.

cable.c Installation

The cable.c program uses the socat Linux utility in the background to create the virtual serial ports in your PC. Therefore, you must install socat to use the cable.c program.

```
sudo apt install -y socat
```

If you are using the project code for the Lab 1 available in Moodle, the cable.c program is located in the cable/ directory. If you are not using the project code, download the program from the code examples section in Moodle.

To use cable.c in your PC, compile the program and **run it in a separate terminal**. Note that the program requires sudo permissions and will ask you to **write your password**.

- If you are using the project code for Lab 1, you can use the Makefile to compile and run cable.c.

make run_cable

- If you are not using the project code, compile the cable.c program with GCC and run it.
- gcc -Wall cable.c -o cable

sudo ./cable

Using cable.c to Send and Receive Bytes in the Same PC

After starting the cable.c program, **leave it running in the background** and watch the bytes being sent and received in the virtual serial ports. The terminal will show the two virtual serial ports that were created, which must be used by the transmitter and the receiver programs.

Typically, the virtual serial ports are:

- Transmitter virtual serial port: /tmp/ttyS10
- Receiver virtual serial port: /tmp/ttyS11

To run the transmitter and the receiver protocol, open two new terminals on your PC: one for the transmitter and another for the receiver. In total, you must have three terminals running in your PC: the transmitter, the receiver and the cable.c.

- Run the cable.c program in the first terminal and leave it running in the background.
- Run the transmitter in the second terminal and use the /tmp/ttyS10 virtual serial port.
- Run the receiver in the third terminal and use the /tmp/ttyS11 virtual serial port.

NOTE: If you are using the project code, remember to change the serial ports in the Makefile to the serial ports that are actually being used, either real or virtual (i.e., change the line TX_SERIAL_PORT = /tmp/ttyS10 to the correct value).

Configuring cable.c Parameters

The cable.c program can be configured to run in different modes that help test the protocol. For example, the program can be configured to disconnect the virtual

serial ports or introduce noise in the connection. These modes help you to test the protocol in different conditions, mimicking the behaviour of real serial ports in the laboratory.

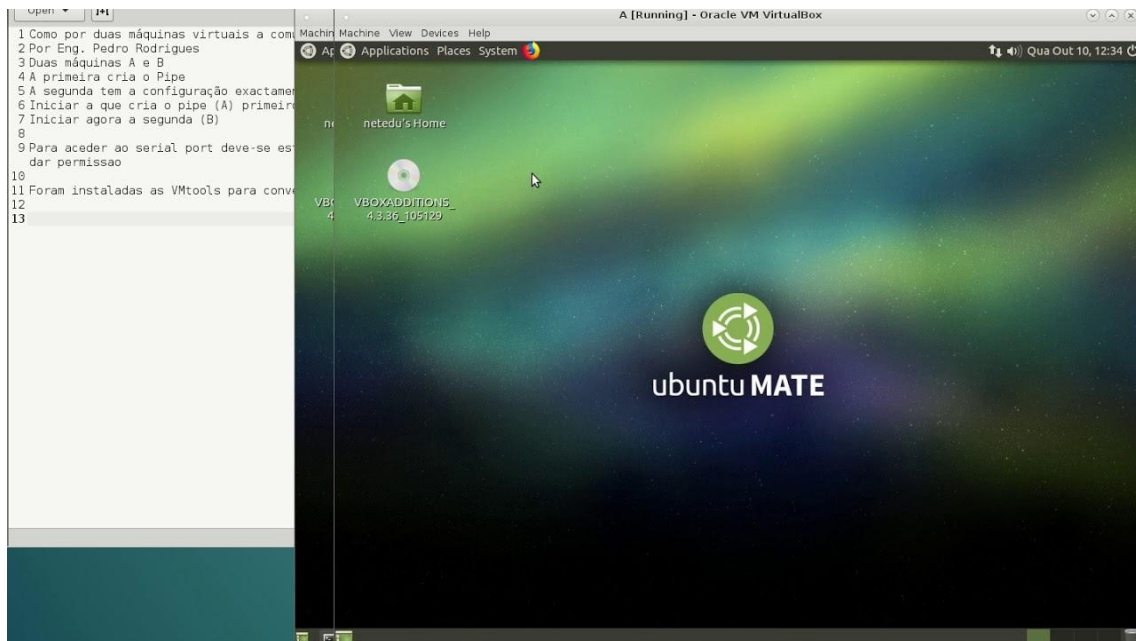
Read the instructions printed in the terminal to configure the cable.c program.

Configure and Use Linux Virtual Machines (Alternative Method)

If you prefer to use virtual machines, you can create two virtual machines on your PC: one for the transmitter and another for the receiver. Note that **this method is generally more complex and requires more configurations**.

To create the virtual machines, follow these steps:

- Install a virtual machine hypervisor, for instance, [VirtualBox](#) or [VMware Player](#).
- Create two virtual machines: one for the transmitter and another for the receiver.
- Configure the virtual serial ports in each virtual machine.
- Connect the virtual serial ports of both virtual machines to each other.
 - If you are using VirtualBox, watch the video tutorial below:



- If you are using VMware, read the [VMware support page](#).